Recall

Generating function.

$$L(x,t) = \frac{1}{\sqrt{1-2xt+t^2}}$$

if $w = 2xt - t^2$

$$L(x,t) = \frac{1}{\sqrt{1-w}} = \sum_{n=0}^{\infty} P_n(x)\, t^n$$

where P_n are the Legendre polynomials.

$$L(x,t) = \sum_{n=0}^{\infty} P_n(x)\, t^n$$

$x = t = \frac{1}{3}$, $L\left(\frac{1}{3},\frac{1}{3}\right) = \dfrac{1}{\sqrt{1 - 2\cdot\frac{1}{3}\cdot\frac{1}{3} + \left(\frac{1}{3}\right)^2}}$

$$= \frac{1}{\sqrt{\frac{9-2+1}{9}}} = \frac{1}{\sqrt{\frac{8}{9}}} = \boxed{\frac{3}{2\sqrt{2}}}$$

$$L\left(\frac{1}{3},\frac{1}{3}\right) = \frac{3}{2\sqrt{2}} = \sum_{n=0}^{\infty} P_n\left(\frac{1}{3}\right)\left(\frac{1}{3}\right)^n$$

$$\boxed{\frac{1}{2\sqrt{2}} = \sum_{n=0}^{\infty} P_n\left(\frac{1}{3}\right)\cdot\frac{1}{3^{n+1}}} \checkmark$$

Q.)

2023-06-06 COMM08 (2) (C2)

We consider the P.D.E

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad , \quad 0 \le x \le 1 \quad , \quad 0 \le y \le \pi.$$

$u(0,y) = 0$ (BC1) $\quad u(1,y) = 0$ (BC2) $\quad u(x,0) = f(x)$ (BC3), $\quad u(x,\pi) = 0$ (BC4)

Sol We assume that our solution $u(x,y)$ is as follows (we use separation of variables)

$$u(x,y) = X(x) \cdot Y(y)$$

After substitution into the PDE we obtain

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} = 0$$

Since x, y are independent $\Rightarrow \exists \lambda \in \mathbb{R}$ such that

$$\frac{1}{X}\frac{d^2X}{dx^2} = \lambda \quad , \quad \frac{1}{Y}\frac{d^2Y}{dy^2} = -\lambda$$

~~Case $\lambda = \alpha^2$~~ $\rightsquigarrow \frac{d^2X}{dx^2} = \alpha^2 X \Rightarrow X(x) = A\cosh(\alpha x) + B\sinh(\alpha x)$

$A = B = 0$

~~Case $\lambda = 0$~~ $\rightarrow \frac{d^2X}{dx^2} = 0 \Rightarrow X(x) = Ax + B$

$A = B = 0$

Case $\lambda = -k^2$

(3)

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$$\frac{d^2X}{dx^2} = -k^2 X$$

$$X(x) = A\cos(kx) + B\sin(kx)$$

BC$_1$ $\Rightarrow$ $A = 0$

BC$_2$ $\Rightarrow$ $k = n\pi$.

$$\boxed{X(x) = B_n \sin(n\pi x), \quad n = 1, 2, 3, \cdots}$$

We solve now, for Y.

$\Rightarrow$

$$\frac{d^2Y}{dy^2} = -\lambda Y = -(k^2)Y = +k^2 Y = n^2\pi^2 Y$$

$$\frac{d^2Y}{dy^2} = n^2\pi^2 Y$$

$$Y(y) = A_n \cosh(n\pi y) + B_n \sinh(n\pi y)$$

BC$_4$ $u(x,\pi) = 0 \Rightarrow$

$$Y(\pi) = 0 = A_n \cosh(n\pi^2) + B_n \sinh(n\pi^2)$$

$$\Rightarrow \boxed{B_n = -A_n \frac{\cosh(n\pi^2)}{\sinh(n\pi^2)}}$$

$$\boxed{Y(y) = A_n \cosh(n\pi y) - A_n \frac{\cosh(n\pi^2)}{\sinh(n\pi^2)} \sinh(n\pi y)}$$

(4)

(4)

$$u(x,y) = X(x) \cdot Y(y)$$

$$u(x,t) = \sum_{n=0}^{\infty} \left(A_n \cosh(n\pi y) - A_n \frac{\cosh(n\pi^2)}{\sinh(n\pi y)} \cdot \sinh(n\pi y) \right) B_n \sin(n\pi x)$$

$$= \sum_{n=0}^{\infty} \underbrace{A_n B_n}_{C_n} \left(\underbrace{\cosh(n\pi y)}_{1} - \frac{\cosh(n\pi^2)}{\sinh(n\pi^2)} \cdot \underbrace{\sinh(n\pi y)}_{0} \right) \sin(n\pi x)$$

Use BC 5

$$u(x,0) = f(x) = \sum_{n=0}^{\infty} C_n (1 - 0) \sin(n\pi x)$$

$$f(x) = \sum_{n=0}^{\infty} C_n \sin(n\pi x)$$

Q: Consider the PDE given by:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + 4\frac{\partial u}{\partial x} + u, \quad 0 \le x \le \pi, \; t \ge 0$$

$$u(t,0) = 0, \quad u(t,\pi) = 0, \quad u(0,x) = f(x)$$

BC_1 BC_2 BC_3

Sol First, we assume that our solution is as follows:

$$u(t,x) = e^{\alpha x + \beta t} \cdot v(t,x)$$

$$\begin{cases} \frac{\partial u}{\partial t} = e^{\alpha x+\beta t}\left(\beta v + \frac{\partial v}{\partial t}\right) \\ \frac{\partial u}{\partial x} = e^{\alpha x+\beta t}\left(\alpha v + \frac{\partial v}{\partial x}\right) \\ \frac{\partial^2 u}{\partial x^2} = e^{\alpha x+\beta t}\left(\alpha^2 v + 2\alpha\frac{\partial v}{\partial x} + \frac{\partial^2 v}{\partial x^2}\right) \end{cases}$$

$$e^{\alpha x+\beta t}\left(\beta v + \frac{\partial v}{\partial t}\right) = e^{\alpha x+\beta t}\left(\alpha^2 v + 2\alpha\frac{\partial v}{\partial x} + \frac{\partial^2 v}{\partial x^2}\right) + 4e^{\alpha x+\beta t}\left(\alpha v + \frac{\partial v}{\partial x}\right) + e^{\alpha x+\beta t} v$$

$$\beta v + \frac{\partial v}{\partial t} = \alpha^2 v + 2\alpha\frac{\partial v}{\partial x} + \frac{\partial^2 v}{\partial x^2} + 4\alpha v + 4\frac{\partial v}{\partial x} + v$$

$$\frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2} + \underbrace{\left(\alpha^2 - \beta + 4\alpha + 1\right)}_{b_0} v + \underbrace{\left(4 + 2\alpha\right)}_{b_1}\frac{\partial v}{\partial x}$$

$4 + 4\alpha = 0 \Rightarrow \boxed{\alpha = -1}$

$(-1)^2 - \beta + 4(-1) + 1 = 0 \Rightarrow \boxed{\beta = -3}$

$4 \qquad -\beta \qquad +1$

$$\Rightarrow \boxed{u(t,x) = e^{-2x-3t} v(t,x)}$$

$$\Rightarrow \frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2}$$

BC$_1$ $\quad u(t,0) = 0 = e^{-2(0)-3t} v(t,0) = e^{-3t} v(t,0)$

$\Rightarrow v(t,0) = 0$ $\quad$ New BC$_1$

BC$_1$ $\quad u(t,\pi) = 0 = e^{-2\pi-3t} v(t,\pi)$

$\Rightarrow v(t,\pi) = 0$ $\quad$ New BC$_2$

BC$_2$ $\quad u(0,x) = f(x) = e^{-2x-3(0)} v(0,x)$

$= e^{-2x} v(0,x)$

$\Rightarrow v(0,x) = e^{2x} f(x)$ $\quad$ New BC$_3$

$$\frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2}$$

$v(t,0) = 0$, $\quad v(t,\pi) = 0$, $\quad v(0,x) = e^{2x} f(x)$

NBC$_1$ $\quad$ NBC$_2$ $\quad$ NBC$_3$

To solve this New Problem
we use Separation of variables

$$u(t,x) = T(t)\,X(x)$$

$$\frac{1}{T}\frac{dT}{dt} = \frac{1}{X}\frac{d^2X}{dx^2}$$

$\Rightarrow$ Separate constant λ.

$$\frac{1}{T}\frac{dT}{dt} = \lambda \quad \& \quad \frac{1}{X}\frac{d^2X}{dx^2} = \lambda$$

$\lambda > 0$

$\lambda = 0$

$\lambda < 0$